

Dynamic State Maintenance and Rule-Constrained Generation in LLM-Based Interactive Narratives

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Abstract

Large Language Models (LLMs) can generate flexible interactive narratives, but maintaining consistent game states and rule-constrained behavior over long-horizon interactions remains challenging. This study investigates whether an external Observer LLM can support an LLM-based Game Master (GM) by decoupling state tracking from narrative generation. The Observer continuously extracts and maintains game-state information from interactions and provides it to the GM. We compare four game-state management architectures and conduct two experiments evaluating terminal-state recognition and rule-constrained interactive narratives. Results show that the Observer substantially improves continuous state tracking and improves termination recognition for some state-dependent conditions, although it does not consistently reduce termination error. It also increases task completion from 20% to 90% under non-compliant player actions. However, these improvements do not consistently enhance narrative coherence or termination detection, while unsolicited actions become more frequent. These findings suggest that external observation can improve state maintenance, but state tracking alone is insufficient to ensure narrative consistency and faithful player interaction.

1 Introduction

LLMs have demonstrated strong capabilities in generating diverse narratives and supporting interactive storytelling. However, maintaining logical and causal consistency across generated narratives remains challenging, as models frequently struggle to track evolving world states and adjudicate rules consistently (Li et al., 2026; Ahuja et al., 2025).

Existing instruction-following benchmarks primarily evaluate adherence to general instructions, while domain-oriented agents must additionally follow external rules that may be complex and dynamically dependent on the environment (Diao et al.,

2025). Relatedly, recent studies have identified limitations in the ability of LLMs to reason consistently about causal relationships in narratives (Yamin et al., 2025).

Recent studies have increasingly explored games as environments for evaluating the capabilities of LLM-based agents (Hu et al., 2025). Even in relatively open-ended environments such as sandbox games, achieving a goal often requires a sequence of state-dependent actions and decisions (Baker et al., 2022).

However, interactive narrative settings introduce an additional challenge: the relevant information is not static. Player actions continuously modify the underlying world state, and subsequent actions must remain consistent with these changes over multiple interactions (Ammanabrolu and Riedl, 2019; Park et al., 2023).

Importantly, game-specific rules do not necessarily coincide with real-world commonsense knowledge. An LLM may possess prior knowledge about how an event should normally occur, while an interactive environment may impose explicit rules that override such expectations. Therefore, a Game Master must not only generate plausible narratives, but also apply environment-specific rules consistently as the world state changes—acting not merely as a narrator, but as an arbiter that determines whether player actions are valid under the current state.

Existing benchmarks have examined narrative consistency, instruction following, and long-horizon agent behavior, but they do not fully capture the combined challenge of maintaining a changing world state while enforcing state-dependent rules and action prerequisites in interactive narratives. This gap is important to address because a Game Master may produce a narratively coherent response while incorrectly updating the world state, inconsistently applying game rules, accepting actions whose prerequisites are not satis-

fied, or failing to terminate the game when a terminal condition is reached.

To investigate these challenges, we develop an evaluation framework for LLM-based Game Masters in interactive narrative environments. We examine termination detection, state-dependent termination, rule consistency during cooperative gameplay, and prerequisite enforcement under invalid player actions. We evaluate multiple Game Master architectures and analyze their ability to maintain consistent world states and enforce environment-specific constraints throughout multi-turn interactions.

2 Related Work

2.1 LLM-based Interactive Narrative

Large Language Models (LLMs) have demonstrated strong capabilities in generating coherent natural-language narratives and flexibly responding to user interactions. In interactive narrative games, acting as a Game Master (GM) requires both interpreting world states and generating appropriate narrative content. Exploring the potential of LLMs as GM assistants, [Zhu et al. \(2023\)](#) showed that LLMs can leverage contextual information to effectively support tabletop role-playing games (TRPGs).

Beyond assisting human GMs, subsequent work has explored the use of LLMs to directly generate and manage interactive narratives. The ability of LLMs to interpret free-form player inputs and produce contextually appropriate responses makes them particularly suitable for open-ended interactions, where players may take actions that are not explicitly anticipated by the game designer. This flexibility allows LLM-based GMs to provide more dynamic and less scripted interactions than traditional rule-based systems ([Peng et al., 2024](#)).

2.2 Narrative Consistency and Causal Reasoning

However, an LLM-based GM is required to do more than generate plausible narrative text. It must also maintain the state of the game world and ensure that generated events and actions remain compatible with previously established information. As the interaction progresses, inconsistencies in world states, character information, or previously established events can accumulate and undermine the coherence of the interactive experience. Thus, maintaining consistency across multiple turns represents a fundamental challenge for LLM-based interactive

narrative systems ([Ma et al., 2026](#)).

[Park et al. \(2026\)](#) systematically categorized narrative consistency in LLM-generated stories into multiple dimensions, including world and setting, character-agentive, event-structural, and narration and discourse consistency. Their taxonomy further identifies different types of consistency failures, providing a structured framework for analyzing inconsistencies in generated narratives. In this study, we adopt this taxonomy as a reference for designing our evaluation criteria and analyzing consistency failures in multi-turn game interactions.

2.3 Rule Following and State Tracking

In interactive narrative games, the Game Master is often required to follow explicit game rules that may differ from the commonsense knowledge encoded in an LLM. Previous studies have shown that LLMs can struggle to consistently follow rules expressed in natural language, even when the rules themselves are relatively simple ([Mu et al., 2024](#)).

Moreover, following rules in interactive environments may require inferential reasoning rather than simply reproducing or following static instructions, as the applicability of a rule can depend on the current state of the environment ([Sun et al., 2024](#)).

For example, a player may initially be unable to open a locked door because they do not possess the required key. However, after acquiring the key, the same action becomes valid. Therefore, the GM must not only remember the relevant rules, but also track changes in the game state and determine whether the corresponding constraints remain applicable.

2.4 External Memory and Structured Knowledge Representation

Recent studies have explored the use of external memory or structured knowledge representation to decouple key information and world states from the generative model, allowing such information to be retrieved and referenced during subsequent interactions.

The Generative Agents framework proposed by [Park et al. \(2023\)](#) demonstrates the application of external memory in interactive environments. By storing interaction histories and environmental information as external memory, LLM-based agents can maintain more consistent behaviors and states throughout continuous interactions.

[Ammanabrolu and Riedl \(2019\)](#) further applied dynamic knowledge graphs to text adventure games

to represent entities, relationships, and states within the game world, enabling models to make subsequent decisions based on structured representations of the world.

These studies demonstrate that extracting key state information from the LLM’s generation process and preserving it through external memory or structured representations can reduce the risk of losing important information during interactions.

However, existing studies have primarily focused on helping agents remember past events or represent the state of the game world, with less attention given to how an independent external observation mechanism can continuously track game-state changes and rule-relevant information to support rule adherence during long-horizon interactions.

2.5 Research Gap

The literature reviewed above highlights several key challenges faced by LLM-based interactive narrative systems.

First, although LLMs offer substantial flexibility in generating and managing open-ended narratives, maintaining causal coherence and consistency in the underlying state of the game world remains challenging. In particular, when explicit game rules are introduced, these rules may conflict with the general commonsense knowledge acquired during pre-training, making it more difficult for the model to apply them accurately.

Second, as rules and environmental states dynamically evolve, the Game Master LLM must continuously track the game state and determine whether a player’s actions comply with the applicable rules.

Third, recent studies have explored external memory and structured representations for preserving interaction histories and world states, suggesting that decoupling state information from the generation process can help mitigate information loss during long-horizon interactions. However, systematic investigations into how LLM-based Game Masters can effectively address these challenges remain limited.

Specifically, this gap can be distilled into two key questions. First, can an independent external observation mechanism continuously and effectively track evolving game states and rule constraints? Second, can providing the Game Master with externally maintained state memory improve its rule adherence and narrative consistency during long-

horizon interactions? Based on these research gaps, this study investigates the following research question:

Research Question: *How does an external Observer LLM for continuous game-state tracking affect state maintenance, rule adherence, task completion, and narrative consistency in long-horizon interactive narratives?*

3 Experiments

To investigate whether an external Observer LLM can improve the reliability of an LLM-based Game Master, we implement a custom narrative simulation platform powered by GPT-4o-mini (OpenAI et al., 2024), which serves as the central Game Master LLM. We conduct two main experimental settings with progressively increasing levels of state and rule complexity.

Experiment 1 (Simulation Environment) focuses on terminal-state recognition, evaluating whether the Game Master can correctly determine when the game should terminate based on the underlying game state. Experiment 2 (Rule-Constrained Interactive Narrative) introduces explicit logical constraints and state-dependent preconditions, requiring the Game Master to determine whether player actions are authorized under the current game state.

We evaluate both experimental settings using four different architectures, enabling us to investigate terminal-state recognition and rule-constrained state tracking in interactive narrative generation.

3.1 Evaluated Frameworks

We evaluate four framework configurations that differ in how game-state updates are maintained and represented. Table 1 summarizes the key architectural differences among these configurations. The first two configurations delegate state maintenance to the Game Master LLM, while the third and fourth configurations externalize state updates to dedicated components.

1. **Standalone LLM (Implicit):** The Game Master LLM is responsible for both narrative generation and implicit state tracking. Numerical state changes are not explicitly provided to the Game Master during the simulation and are only available when the game terminates.

Table 1: Framework Comparison

Framework	State Update	State Representation	External Component
Implicit LLM	Game Master LLM	Hidden	None
Explicit LLM	Game Master LLM	Explicit	None
Programmatic	Python	External	Python Tracker
Observer LLM	Observer LLM	External	Observer LLM

2. **Standalone LLM (Explicit):** Similar to the first framework, the Game Master LLM is responsible for narrative generation and state tracking. However, state tracking is performed explicitly. The current game state and the specific rules applied are displayed to the Game Master at the end of each turn.
3. **Deterministic Programmatic State Tracker:** The Game Master LLM focuses solely on narrative generation, while state updates are calculated deterministically using Python code based on the actions performed during each turn. The resulting game state is then provided to the Game Master LLM to support subsequent narrative generation.
4. **Collaborative Framework with an Auxiliary Observer LLM:** Following the Game Master’s narrative generation, the dialogue history between the player and the Game Master is passed to an auxiliary Observer LLM. The Observer extracts and updates the game state based on the observed interaction and provides the resulting state information to the Game Master LLM to support subsequent narrative progression.

These configurations provide progressively different mechanisms for maintaining game state, ranging from implicit state tracking within the Game Master to external state tracking performed by a deterministic program or an independent Observer LLM.

3.2 Experiment 1: Simulation Environment

We implement a controlled interactive narrative environment based on a spacecraft survival scenario. Following a spacecraft accident, the player must survive in an abandoned space mining station and ultimately transmit a distress signal to complete the game. The simulation is governed by three core state variables: power, oxygen, and stress. These

variables dynamically change as the player interacts with the environment. When the state reaches one of two terminal conditions—resource depletion (i.e., when either power or oxygen drops to zero, indicating total failure of life support systems) or psychological collapse (i.e., when stress exceeds a threshold of 10 points)—the game ends.

3.2.1 Setting: Terminal-State Recognition

The first experiment evaluates whether the Game Master can correctly recognize terminal conditions based on the current game state. The Game Master interacts with the player until a predefined terminal condition is reached. Once the relevant state reaches its termination threshold, the Game Master is expected to correctly identify the terminal state and end the game.

The experiment evaluates whether the Game Master can distinguish between non-terminal and terminal states and correctly determine when the game should terminate.

3.2.2 Setting: Comparison Across Terminal-State Types

This setting further examines whether terminal-state recognition varies depending on the type of state responsible for termination. Three terminal-state conditions are independently evaluated: power depletion, oxygen depletion, and pressure reaching its critical threshold.

By comparing these three conditions, we investigate whether the Game Master’s ability to recognize game termination is affected by the type of underlying state or termination condition.

3.3 Experiment 2: Rule-Constrained Interactive Narrative

The second experiment extends the simulation by introducing explicit logical constraints and state-dependent preconditions. Unlike Experiment 1 (Simulation Environment), which primarily focuses on recognizing terminal states, Experiment 2 (Rule-Constrained Interactive Narrative) requires

the Game Master to determine whether a player’s action is valid under the current game state.

In this experiment, the setting remains the same abandoned space mining station as in Experiment 1 (Simulation Environment). However, certain actions can only be performed after their corresponding preconditions have been satisfied. For example, a player may need to acquire a required item or complete a specific action before being authorized to perform a subsequent action. Therefore, the Game Master must track changes in the game state and apply the corresponding rules when interpreting each player’s action.

The final objective remains to successfully transmit a distress signal. The game terminates once this objective is successfully completed.

3.3.1 Setting: Compliant Player Actions

In the Compliant Player Actions setting, the player follows the predefined game rules and performs actions that satisfy the required preconditions. The player progressively completes the required actions and ultimately transmits the distress signal.

This setting evaluates whether the Game Master can correctly maintain the evolving game state and apply state-dependent rules when the player follows the intended progression of the game.

3.3.2 Setting: Non-Compliant Player Actions

In the Non-Compliant Player Actions setting, the player intentionally performs actions that violate predefined preconditions or attempts to perform actions before the required conditions have been satisfied. The Game Master must determine whether each action is authorized according to the current game state and respond consistently with the applicable rules.

This setting evaluates the robustness of rule adherence under invalid player actions. In addition to state-tracking accuracy, we evaluate rule violations, unauthorized actions, and textual logical consistency to determine whether the Game Master’s generated narrative remains consistent with the current state and applicable constraints.

4 Experiments and Evaluation

4.1 Experimental Setup

Each player dialogue input represents one discrete simulation turn. The game state is updated deterministically according to a predefined set of consumption rules.

4.1.1 Predefined Consumption Rules

The simulation maintains three core state variables: power, oxygen, and stress. Their values are updated after each player action according to the following deterministic rules:

1. **Power:** Each turn incurs a baseline cost of -2. Specific actions induce additional costs, including area illumination (-1), operating a plasma welder (-10), and transmitting a distress signal (-5).
2. **Oxygen:** Each turn incurs a baseline cost of -1. Exploratory actions induce an additional cost of -2.
3. **Stress:** Each turn incurs a baseline accumulation of +1. Actions performed in dark environments induce an additional accumulation of +2.

4.2 Experiment 1: Terminal-State Recognition

4.2.1 Experimental Setting

The initial game state is defined as 40 units of power, 20 units of oxygen, and 0 units of stress. The game terminates when power falls below 0, oxygen falls below 0, or stress exceeds 10.

Each trial consists of a maximum of 20 player actions. If the Game Master does not terminate the game after 20 actions, the trial is considered a termination failure.

Based on the deterministic state-transition rules, the corresponding terminal conditions are expected to be reached at different stages of the simulation. Under the evaluated state trajectories, stress exceeds its threshold at approximately the ninth turn, power falls below its threshold at approximately the tenth turn, and oxygen falls below its threshold at approximately the fifteenth turn.

We evaluated four Game Master architectures: Implicit LLM, Explicit LLM, programmatic, and Observer LLM. Each architecture is evaluated in 10 independent trials.

4.2.2 Terminal-State Recognition Results

The Termination Detection Rate measures the proportion of trials in which the game is terminated within the 20-turn limit. Termination Correctness measures whether the termination occurred at a state that satisfies either the predefined terminal condition or the accepted boundary-state criterion. The predefined terminal conditions are

Table 2: Terminal-State Recognition Results

Metric	Implicit LLM	Explicit LLM	Programmatic	Observer LLM
Terminated Trials	3/10	10/10	6/10	10/10
Termination Detection Rate	30%	100%	60%	100%
Correctly Terminated Trials	0/3	10/10	6/6	10/10
Termination Correctness	0%	100%	100%	100%
Mean Termination Error	N/A	0.6	2.83	3.3

power < 0 , oxygen < 0 , and stress > 10 . For evaluation purposes, we also accept boundary states in which power or oxygen reaches exactly 0, or stress reaches exactly 10, as valid termination points. These boundary states occur immediately before the predefined terminal conditions are reached and are therefore considered acceptable, but they are assigned a termination error of one turn to reflect the difference between the observed termination point and the expected termination point under the predefined rules.

For each trial in which termination occurred, we further measure the Termination Error as the absolute difference between the actual termination turn and the expected termination turn. The Mean Termination Error is calculated over all trials in which termination was detected. The results are shown in Table 2.

Analysis: As shown in Table 2, the Implicit LLM terminated the game in only 3 out of 10 trials, and none of these terminations corresponded to a valid terminal state. In contrast, the Explicit LLM terminated the game in all 10 trials, with all 10 terminations satisfying the terminal-state criteria. It also achieved the lowest mean termination error among the evaluated architectures, with an average error of 0.6 turns. The Programmatic and Observer LLM architectures also achieved correct termination in all trials in which termination was detected, although the Observer LLM exhibited a higher mean termination error of 3.3 turns.

4.2.3 Terminal-State Recognition Ablation Study

In the initial terminal-state recognition experiment, the Explicit LLM achieved the lowest mean termination error. To examine whether explicitly presenting state information contributed to this performance, we conducted an additional ablation study. Specifically, we modified the Programmatic and Observer LLM configurations to append the same rule calculations and current game-state informa-

tion used in the Explicit LLM configuration to the end of each Game Master’s response.

Both configurations showed improved termination performance after incorporating this additional information. However, this modification introduces an additional explicit state representation that was not part of the original design of these frameworks. In particular, providing the same explicit state information to the Programmatic and Observer LLM configurations changes the intended architectural distinction between the evaluated frameworks.

Therefore, we do not include this modified configuration in the subsequent experiments. Instead, the original framework configurations are retained to preserve a consistent comparison of their respective state-tracking mechanisms.

4.2.4 Comparison Across Terminal-State Types

To further examine whether terminal-state recognition varies across different types of terminal conditions, we conduct a controlled comparison of power, oxygen, and stress termination.

For each condition, the initial game state is configured to be close to the corresponding termination threshold, such that the deterministic state-transition rules lead to termination within approximately three turns. Each trial allows a maximum of six player actions.

For the power-termination condition, the initial state is set to 5 units of power, 20 units of oxygen, and 0 units of stress. For the oxygen-termination condition, the initial state is set to 10 units of power, 2 units of oxygen, and 0 units of stress. For the stress-termination condition, the initial state is set to 10 units of power, 10 units of oxygen, and 9 units of stress Table 3.

Each condition is evaluated using five independent trials for both the Explicit LLM and Observer LLM architectures. The results are presented in Table 4.

Condition	Initial	Expected termination
Power	5	Turn3
Oxygen	2	Turn3
Stress	9	Turn2

Table 3: Comparison Across Terminal-State Types

4.3 Experiment 2: Rule-Constrained Interactive Narrative

Experiment 2(Rule-Constrained Interactive Narrative) extends the game environment used in Experiment 1(Simulation Environment) by introducing explicit action preconditions and a sequential task structure. At the beginning of the game, the mining station has only emergency power. To restore the station’s power supply, the player must first retrieve diesel fuel from the storage room and then add the fuel to the generator in the engineering room. After the generator is activated, the gate to the control center can be opened, allowing the player to access the communication system. The player must then use the communication system to transmit a distress signal and complete the game.

Each action in this sequence is subject to its corresponding preconditions. For example, the player cannot activate the generator without first obtaining the required diesel fuel, and the control center cannot be accessed before the generator has been activated. If a player attempts to perform an action before its preconditions are satisfied, the Game Master should prevent the action from being executed and explain the reason to the player.

4.4 Evaluation Metrics

Experiment 2(Rule-Constrained Interactive Narrative) consists of two experimental conditions: Compliant Player Actions and Non-Compliant Player Actions. Each condition is evaluated using 10 independent trials, with each trial consisting of 10 dialogue turns.

We evaluate 12 metrics in total, divided into trial-level metrics and turn-level metrics. Trial-level metrics characterize the overall outcome of each game, whereas turn-level metrics capture errors and behaviors occurring during individual dialogue turns.

Trial-Level Metrics:

1. **Task Completion:** Whether the player successfully completes the final task objective of transmitting a distress signal.

2. **Termination Detection:** Whether the Game Master correctly terminates the game when the task has been successfully completed.
3. **First Error Turn:** The first dialogue turn in which the Game Master produces an incorrect, inconsistent, or rule-violating response.
4. **Narrative Coherence:** Whether the overall narrative remains logically coherent throughout the trial.

Turn-Level Metrics:

1. **Game-State Hallucination:** Whether the Game Master introduces an unsupported game element or state that is not defined by the original game rules and affects the progression of the game.
2. **State Tracking Accuracy:** Whether the game-state variables are consistent with the actual game state. The tracked variables include **Continuous Numerical States**, such as power, oxygen, and stress, as well as discrete Boolean states representing task and game progress. This metric consists of two components: **State Tracking Accuracy**, which measures whether the game-state variables are correctly maintained at each dialogue turn, and **State Transition Accuracy**, which measures whether changes in game states are correctly tracked when state transitions occur.
3. **Unsolicited Action:** Whether the Game Master causes the player character to perform an action that was not requested by the player.
4. **Rule Violation:** Whether the Game Master allows or describes an action that violates one or more predefined action preconditions.
5. **Action Execution Failure:** Whether the Game Master’s response fails to describe or acknowledge an action explicitly requested by the player.
6. **Action Blocking:** Whether the Game Master correctly prevents a requested action whose preconditions have not been satisfied and provides an appropriate explanation for why the action cannot be performed.

Table 4: Terminal-State Types Results

Terminal Condition	Framework	Termination Detection Rate	Mean Termination Error
Power	Explicit LLM	5/5	0.6
Power	Observer LLM	5/5	0.2
Oxygen	Explicit LLM	5/5	1.0
Oxygen	Observer LLM	5/5	1.0
Stress	Explicit LLM	1/5	0
Stress	Observer LLM	5/5	3.2

4.5 Rule-Constrained Interactive Narrative results

The results of Experiment 2 are summarized in Table 5. The evaluation compares the Explicit LLM and Observer LLM under Compliant and Non-Compliant Player Actions, focusing on task completion, state tracking, and rule-constrained behavior.

4.5.1 Task Completion and Termination

Under the Compliant Player Actions condition, the Explicit LLM completed the task in 5 out of 10 trials, while the Observer LLM completed it in 7 out of 10 trials. Under the Non-Compliant Player Actions condition, the difference was substantially larger: task completion increased from 2/10 (20%) with the Explicit LLM to 9/10 (90%) with the Observer LLM.

In compliant trials, failures were primarily caused by the Game Master incorrectly reporting that the generator failed to activate, preventing the player from restoring power and accessing the communication system. In non-compliant trials, the Game Master sometimes described the player as completing prerequisites that had not been requested. Although such behavior deviated from the player’s requested actions, it allowed the game to progress and ultimately complete the task.

Termination detection was less consistent across the two architectures. Among successfully completed trials, the Observer LLM did not consistently outperform the Explicit LLM in detecting the terminal state. This suggests that the improvement in task completion does not necessarily translate into improved termination detection.

4.5.2 State Tracking

State tracking was evaluated using Continuous Numerical States, State Tracking Accuracy, and State Transition Accuracy. Continuous Numerical States measure whether changes in power, oxygen, and

stress were correctly tracked, while State Tracking Accuracy measures the correctness of discrete Boolean states representing task and game progress. State Transition Accuracy measures whether these Boolean states were correctly updated when state changes occurred.

The most notable difference was observed in continuous numerical state tracking. The Explicit LLM correctly tracked only 10/300 numerical state updates under the Compliant condition and 13/300 under the Non-Compliant condition. In contrast, the Observer LLM correctly tracked 174/291 and 152/291 updates, respectively. This indicates that the Observer LLM substantially improved the tracking of continuously changing numerical game states.

The Observer LLM completed only seven dialogue turns in its tenth trial under the corresponding condition, as the Game Master prematurely terminated the game by having the player transmit the distress signal. Consequently, the number of evaluable state updates was lower for the Observer LLM.

For discrete Boolean states, both architectures achieved very high State Tracking Accuracy. The Explicit LLM achieved 698/700 and 700/700 under the Compliant and Non-Compliant conditions, respectively, while the Observer LLM achieved 672/679 and 675/679. Thus, the Observer LLM did not improve overall Boolean state-tracking accuracy. State Transition Accuracy was also generally high for both architectures, with the Observer LLM performing slightly better under the Compliant condition but slightly lower under the Non-Compliant condition.

4.5.3 Rule-Constrained Behavior

Rule violations were rare in both architectures, with only one violation observed for each architecture under the Compliant condition and none under the Non-Compliant condition. Action Execution Failure was similarly rare.

Table 5: Rule-Constrained Interactive Narrative Results

Metric	Explicit LLM		Observer LLM	
	Compliant	Non-Comp.	Compliant	Non-Comp.
Task Completion	5/10	2/10	7/10	9/10
Termination Detection	1/5	2/2	1/7	6/9
First Error Turn	2.5	2.7	3.5	2.0
Narrative Coherence	10/10	9/10	8/10	8/10
Game-State Hallucination	1/100	2/100	2/93	1/93
Continuous numerical states	10/300	13/300	174/291	152/291
State Tracking Accuracy	698/700	700/700	672/679	675/679
State Transition Accuracy	42/44	30/30	51/52	61/63
Unsolicited Action	1/100	22/100	15/93	32/93
Rule Violation	1/100	0/100	1/93	0/93
Action Execution Failure	0/100	1/100	0/93	1/93
Action Blocking	10/100	26/100	4/93	20/93

The denominator is smaller than 300 for the Observer LLM because one trial terminated prematurely after seven turns, resulting in 291 evaluable state updates.

However, the Observer LLM produced more Unsolicited Actions than the Explicit LLM. Under the Compliant condition, unsolicited actions occurred in 15/93 turns for the Observer LLM compared with 1/100 for the Explicit LLM. Under the Non-Compliant condition, the corresponding rates were 32/93 and 22/100. This behavior is consistent with the observed cases in which the Observer LLM introduced additional actions, such as forcibly moving the player to satisfy task prerequisites.

Overall, the Observer LLM did not produce a general improvement across all metrics. Its most notable benefits were continuous numerical state tracking and task completion under Non-Compliant Player Actions, where task completion increased from 20% to 90%. However, this improvement was accompanied by more unsolicited actions, suggesting a potential trade-off between maintaining task progression and strictly following the player’s requested actions.

5 Conclusion & Future Work

5.1 Discussion

External observation improves certain aspects of state maintenance, but state tracking alone does not guarantee globally consistent or player-faithful behavior.

5.1.1 Observer LLM as External State Support

The results of the terminal-state recognition experiments indicate that the Observer LLM can assist the Game Master LLM in recognizing state-dependent termination conditions. In the Terminal-State Recognition setting, the Explicit LLM cor-

rectly identified the termination point in all trials. However, in the Comparison Across Terminal-State Types setting, the Explicit LLM failed to correctly identify the termination condition caused by Stress, whereas the Observer LLM correctly identified all tested types of terminal conditions. These results suggest that externally maintained state information can support the Game Master in recognizing termination conditions that depend on continuously changing game states.

The results from the Rule-Constrained Interactive Narrative experiment further support this observation. The Observer LLM achieved substantially higher accuracy in tracking Continuous Numerical States, including Power, Oxygen, and Stress, than the Explicit LLM. In particular, the Explicit LLM frequently retained previously generated numerical values without consistently applying state changes as the dialogue progressed. In contrast, the Observer LLM was able to maintain a substantially larger proportion of the numerical state updates. This suggests that external observation can reduce the burden of continuously maintaining numerical game states within the narrative generation process.

5.1.2 State Tracking Does Not Guarantee Narrative Consistency

Although the Observer LLM improved the maintenance of numerical game states, this improvement did not translate into better overall narrative coherence. The Observer LLM achieved lower Narrative Coherence scores than the Explicit LLM in both experimental conditions. Examination of the generated interactions suggests that this behavior may be related to the Observer LLM’s tendency to ag-

gressively satisfy action prerequisites in order to maintain game progression.

For example, in one trial, the player searched for diesel and obtained a tightly closed diesel container that could not be opened using the items currently held by the player. However, when the player subsequently attempted to add the diesel to the generator, the Game Master described the diesel container as being opened without providing a corresponding action or explanation. This allowed the required state transition to occur, but introduced an inconsistency between the previously established narrative state and the subsequent action.

A similar behavior was observed in the final trial of the Observer LLM under both experimental settings. The experimental design specified that the player would request to transmit the distress signal only in the tenth dialogue turn. However, the Observer LLM caused the player to activate the generator before the requested turn and subsequently described the player as entering the control center and transmitting the distress signal. Although these actions allowed the game to reach the final objective, they were not requested by the player and altered the intended progression of the interaction.

These observations indicate that maintaining external game-state information is not sufficient to guarantee narrative consistency. The Game Master must also correctly integrate the maintained state with previously established events, action constraints, and player intentions when generating subsequent responses.

5.1.3 State Consistency and Player Agency

Although the Observer LLM substantially improved task completion, it also noticeably increased the frequency of Unsolicited Actions, in which the Game Master performed or described actions that were not requested by the player. One possible explanation is that the way the Observer LLM summarizes and communicates game-state information to the Game Master may cause the Game Master to interpret state descriptions as intended player actions. For example, repeatedly presenting the current game state and required task progress may implicitly encourage the Game Master to advance the player toward the next prerequisite, even when the player has not explicitly requested such an action.

This observation suggests that the design of the interface between the Observer LLM and the Game Master may be an important factor in maintaining

both state consistency and player agency. Future work could investigate this issue through ablation studies that vary the content and format of Observer outputs, as well as through different game structures and task designs. Such experiments could help determine how external state information can be provided to the Game Master effectively without unintentionally influencing its narrative generation or causing it to perform actions on behalf of the player.

5.2 Limitations

This study has several limitations. First, the experiments were conducted using a single narrative scenario based on a mining-station environment. Therefore, the observed results may not generalize to interactive narratives with different settings, task structures, or rule configurations.

Second, the evaluation of the Rule-Constrained Interactive Narrative experiment was conducted manually by the researchers. Although the defined evaluation metrics provide a structured basis for analysis, manual annotation may introduce subjectivity and limits the scalability of the evaluation process. Developing reliable automated evaluation methods for these metrics remains an important challenge.

Third, the Observer LLM itself is not a deterministic state tracker. Because game states are extracted and updated by an LLM, errors in state interpretation or state updating may be introduced by the Observer. Therefore, the external state maintained by the Observer cannot be assumed to be perfectly accurate and may itself affect the subsequent behavior of the Game Master.

5.3 Future Work

Building upon these findings, we intend to pursue several directions for future research.

1. **Diversifying Narrative Settings:** The simulation platform can be extended to different narrative settings, task structures, and rule configurations, while also evaluating different LLMs, to examine the generalizability of the Observer-based framework.
2. **LLM-based Players and Automated Evaluation:** Scripted player actions can be replaced with an LLM-based player to enable more dynamic and diverse interactions. Combined with automated evaluation methods, this would enable a fully automated end-to-end

evaluation loop for long-horizon interactive narratives.

3. **Longer-Horizon and Robustness Evaluation:** Future experiments can extend the interaction horizon beyond 20 dialogue turns and introduce more frequent or dynamic rule and state changes. This would allow us to investigate whether the state-tracking benefits of the Observer framework remain robust over longer and more complex interactions.
4. **Observer–Game Master Communication Design:** Future work can further investigate how state information should be summarized and communicated from the Observer LLM to the Game Master. Ablation studies can compare different state representations and output formats to determine whether the information provided by the Observer can be used effectively without unintentionally influencing the Game Master’s narrative generation or causing unsolicited actions.

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